

D π D Maths

TMUA

Mock Set 1

Paper 1

20 Multiple Choice Questions

Time allowed: 75 minutes

Calculators are not permitted

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Question 1

Let k be a real number. The equation $|x^2 + 3kx| = 2k|x - 3|$ has n solutions for a given k . Find the sum of all values of n in $\{0, 1, 2, 3, 4, 5\}$ that are possible.

- A) 6
- B) 9
- C) 10
- D) 12
- E) 14
- F) 15

Question 2

Find the complete set of x for which $\frac{1}{\sqrt{6 - \log_3(x^4)}}$ is defined.

- A)** $-27 < x < 27, x \neq 0$
- B)** $-3\sqrt{3} < x < 3\sqrt{3}, x \neq 0$
- C)** $-9 < x < 9, x \neq 0$
- D)** $0 < |x| < 27$
- E)** $-3 < x < 3, x \neq 0$
- F)** $|x| \geq 27$
- G)** $-27 \leq x \leq 27$
- H)** $-3\sqrt{3} \leq x \leq 3\sqrt{3}$

Question 3

A circle of radius 7 is drawn centred at the origin. How many squares of side length 1 and integer coordinate vertices intersect the interior of this circle?

- A) 156
- B) 164
- C) 168
- D) 172
- E) 176
- F) 180
- G) 184
- H) 196

Question 4

Let x be a real number. Find the maximum value of 3^{6x-2x^2-5} .

A) $\frac{\sqrt{3}}{3}$

B) $\sqrt{3}$

C) 3

D) $\frac{1}{3}$

E) $\frac{1}{9}$

F) $\frac{\sqrt{2}}{2}$

G) 1

H) $3\sqrt{3}$

Question 5

Define $P(x) = (x-1^2)(x-2^2)\cdots(x-100^2)$. How many integers n are there such that $P(n) \leq 0$?

- A) 4900
- B) 4950
- C) 5000
- D) 5050
- E) 5100

Question 6

Find the constant term of $\left(|x| + \frac{1}{|x|} - 3\right)^3$.

- A) 20
- B) -30
- C) -45
- D) -27
- E) 45
- F) 18
- G) -18
- H) 27

Question 7

Real numbers a, b, c satisfy $a + b + c = 0$ with $0 < a < c$. Find the number of solutions to $c \sin^2 \theta + b \sin \theta + a = 0$ in $0 < \theta < \pi$.

- A) 3
- B) 1
- C) 2
- D) 4
- E) 0
- F) 5
- G) 6
- H) Infinite

Question 8

The ellipse is $\frac{x^2}{9} + y^2 = 1$. Given point $M(2, 0)$ and point P on the ellipse, find the minimum value of $|PM|$.

- A) $\frac{\sqrt{2}}{3}$
- B) $\frac{\sqrt{3}}{3}$
- C) $\frac{\sqrt{6}}{3}$
- D) $\frac{2\sqrt{2}}{3}$
- E) $\frac{\sqrt{2}}{2}$
- F) 1
- G) $\frac{1}{3}$
- H) $\frac{1}{2}$

Question 9

Given $f(x) = |x^2 - 10x + 13|$ and $g(x) = 6x + \frac{6}{x}$. With constant $m \in \mathbb{R}$, let $\epsilon = f(x) - g(m)$, where $\epsilon^2 + 12\epsilon = 0$. Find the maximum number of solutions of x .

- A) 1
- B) 2
- C) 3
- D) 4
- E) 5
- F) 6
- G) 7
- H) 8

Question 10

Let $a_n = \frac{n \cdot 2^{n+1}}{(n+1)(n+2)}$, where n is a positive integer. Find $a_1 + a_2 + a_3 + \cdots + a_{254}$.

- A) $\frac{2^{256}}{256}$
- B) $\frac{2^{256}}{256} - 2$
- C) -2
- D) $\frac{2^{256}}{256} + 2$
- E) 2
- F) $\frac{2^{255}}{255} - 2$
- G) $\frac{2^{256}}{255} - 2$
- H) $\frac{2^{254}}{256} - 2$

Question 11

A curve $y = x^2 - 4x$ is rotated 180° about the origin, then translated by $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$. Find the equation of the new curve.

A) $y = 7 - x^2$

B) $y = 1 - x^2$

C) $y = -1 - x^2$

D) $y = 3 - x^2$

E) $y = 7 - (x - 2)^2$

F) $y = 7 + x^2$

G) $y = -7 - x^2$

H) $y = 4 - x^2$

Question 12

Suppose x and y are positive reals such that $x - y^2 = 1$ and $x^2 + y^4 = 41$. Find x .

A) 4

B) 5

C) $\sqrt{41}$

D) 6

E) $\frac{-1 + \sqrt{17}}{2}$

F) 3

G) 20

H) 9

Question 13

A rectangular sheet of paper is folded along its diagonal, making a pentagon that has an area of $\frac{11}{16}$ of the area of the original rectangle. Find the ratio of the longer side of the rectangle to the shorter side.

- A) $\frac{11}{16}$
- B) $\sqrt{3}$
- C) 2
- D) $\frac{5}{2}$
- E) $\sqrt{5}$
- F) $\frac{3}{2}$
- G) 4
- H) $\sqrt{2}$

Question 14

Tom needs to pay £100 for his dinner. He has an unlimited supply of £2 coins and £5 and £10 notes. In how many ways can he pay?

- A) 60
- B) 63
- C) 66
- D) 55
- E) 121
- F) 36
- G) 72
- H) 100

Question 15

What are the last 6 digits of $11 \times 101 \times 1001 \times 10001 \times 11$?

- A) 344321
- B) 443221
- C) 222111
- D) 443321
- E) 344421
- F) 421443
- G) 434221
- H) 442321

Question 16

Let O be the point $(0, 0)$. Let A, B, C be three points in the plane such that $OA = 30$, $OB = 30$, and $OC = 14$, chosen so that the area of triangle ABC is maximal. What is the length of the shortest side of ABC ?

- A) 44
- B) 40
- C) 48
- D) 34
- E) 30
- F) 42
- G) 38
- H) 36

Question 17

Alice and Bob repeatedly flip a fair coin. Alice wins if 1 head followed by 3 tails appears. Bob wins if 5 heads in a row appears. They flip until someone wins. What is the probability that Alice wins?

- A) $\frac{1}{2}$
- B) $\frac{15}{19}$
- C) $\frac{7}{11}$
- D) $\frac{3}{5}$
- E) $\frac{15}{16}$
- F) $\frac{8}{23}$
- G) $\frac{16}{31}$
- H) $\frac{31}{47}$

Question 18

Let $p(x)$ and $q(x)$ be two cubic polynomials such that $p(0) = -6$, $q(0) = 10$, and $p(q(x)) = q(p(x))$ for all real numbers x . Find the ordered pair $(p(2), q(4))$.

- A)** $(2, -6)$
- B)** $(-6, 2)$
- C)** $(2, 6)$
- D)** $(-6, 10)$
- E)** $(6, -6)$
- F)** $(2, -10)$
- G)** $(-2, 6)$
- H)** $(10, -6)$

Question 19

An ant starts at the origin, facing in the positive x -direction. Each second, it moves 1 unit forward, then turns counterclockwise by $\sin^{-1}\left(\frac{5}{13}\right)$ degrees. What is the least upper bound on the distance between the ant and the origin?

A) $\sqrt{13}$

B) $\sqrt{26}$

C) $\frac{13}{5}$

D) 5

E) $\frac{13}{2}$

F) $\sqrt{10}$

G) $2\sqrt{13}$

H) $\frac{26}{5}$

Question 20

Ava and Ben play a game on a 12×12 board. For some positive integer k , Ava chooses k squares and writes an A in each of them. Ben then chooses $k + 3$ squares and writes a B in each. After Ben is done, Ava wins if there is a sequence of letters forming “ BAA ” or “ AAB ” when read either horizontally (left-to-right) or vertically (top-to-bottom), and Ben wins otherwise. Determine the smallest value of k for which Ava has a winning strategy.

- A) 24
- B) 34
- C) 35
- D) 36
- E) 37
- F) 40
- G) 48
- H) 18